PROBLEM 48

# Two Connected Particles, One on Paraboloid

Original statement · Problems.pdf, p. 33

Multi-Particle Manifold

Two heavy particles $P$ and $Q$, each of mass $m$, are connected with an inextensible, massless string of length $\ell$. Particle $P$ is constrained to stay on a paraboloid, given implicitly by equation $z = cR^2$, where $c$ is a known constant and we have introduced the usual cylindrical coordinates $\{R, \theta, z\}$. The string passes through a small hole at the origin $O$, and particle $Q$ is restricted to hang along the vertical axis. The reference frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial. There is no friction in this problem. Assume that the string is long enough so that $Q$ never reaches $O$ and that it is always tense.

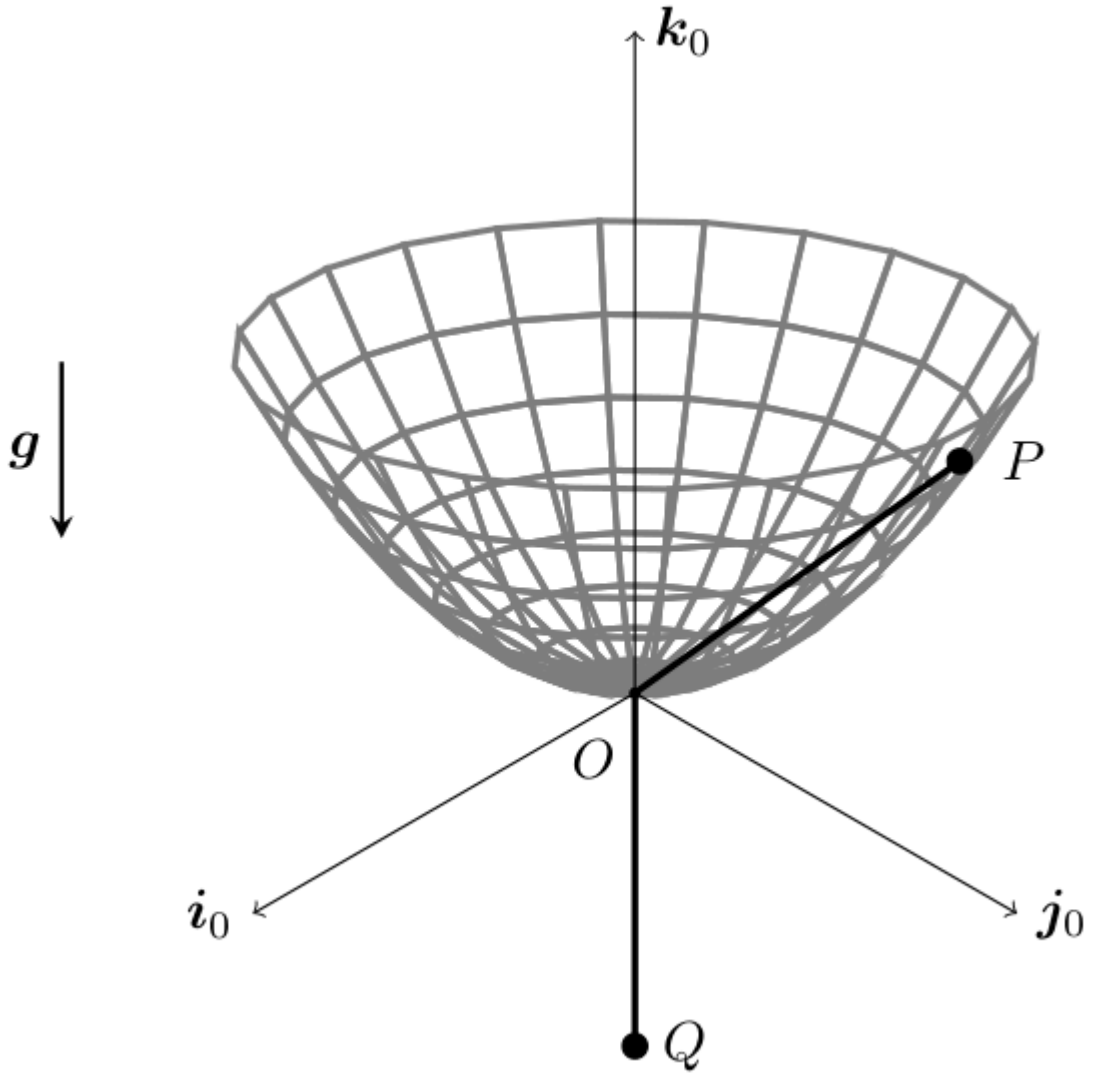



*Figure: Paraboloid with P on its surface, string through the hole at O and Q hanging along the vertical axis k0 (Problems.pdf, p. 33).*

**(a)** How many effective degrees of freedom does the system have? Justify your answer.

**(b)** Give the expressions for the position, velocity, and acceleration of $P$ and $Q$ with respect to $S_0$ as a function of $R, \theta$ and their derivatives. If you use any vector basis other than $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ to write your solution, define it first.

**(c)** Find a normal vector and two linearly independent tangent vectors to the paraboloid as a function of $R, \theta$. They do not need to be unitary.

**(d)** Identify and write down the general expression of all the forces acting on $P$ and $Q$, introducing new unknowns as needed.

**(e)** Find the condition that the solution $R(t)$ must satisfy so that the string is indeed always in tension.

**(f)** Using $S_0$ for your analysis, write down two second-order differential equations for $R$ and $\theta$, that do not depend on any other unknowns.

**(g)** Find two scalar conservation laws (i.e., first integrals).

**(h)** Assuming that initially $R = 1/c, \theta = 0, \dot{R} = 0, \dot{\theta} = d$, where $d \neq 0$ is a given value, write down an equation in the form $f(R) = 0$ whose roots determine the minimum and maximum values of $R$ in the subsequent motion. You do not need to solve this equation.

**Full Analytical Solution (4-Phase Methodology)**

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **Physical System:** Two particles $P$ and $Q$ of mass $m$ connected by an inextensible massless cord of length $\ell$.
- Particle $P$ is constrained to the frictionless paraboloid $z = cR^2$ ($c > 0$).
- Particle $Q$ hangs along the vertical axis $Oz$.
- Cord passes through a frictionless hole at origin $O$.
- **CDOFs:** 2 degrees of freedom: $(R, \theta)$.
- **Initial State:** $R(0) = 1/c, \theta(0) = 0, \dot{R}(0) = 0, \dot{\theta}(0) = d \neq 0$.
- **Parameters:** $m, c, \ell, g, d$.

PHASE 2

**Reference Frames, Vector Bases & Coordinate Geometry**

- Cylindrical basis: $\{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{k}\}$.
- Position of $P$: $\mathbf{r}_P = R\mathbf{e}_R + cR^2\mathbf{k}$.
- Tangent vectors to paraboloid:

$$\mathbf{t}_R = \frac{\partial \mathbf{r}_P}{\partial R} = \mathbf{e}_R + 2cR\,\mathbf{k}$$

$$\mathbf{t}_\theta = \frac{\partial \mathbf{r}_P}{\partial \theta} = R\mathbf{e}_\theta$$

- Normal vector to paraboloid:

$$\mathbf{n} = \mathbf{t}_R \times \mathbf{t}_\theta = (\mathbf{e}_R + 2cR\,\mathbf{k}) \times (R\mathbf{e}_\theta) = R\mathbf{k} - 2cR^2\,\mathbf{e}_R = R(-2cR\,\mathbf{e}_R + \mathbf{k})$$

- Position of $Q$: Distance along cord from $O$ to $P$ along generator:

Cord length constraint: $s_P + s_Q = \ell$.

For small hole at origin with straight cord to $P$: $|\mathbf{r}_P| = \sqrt{R^2 + c^2R^4} = R\sqrt{1 + c^2R^2}$.

Position of $Q$: $\mathbf{r}_Q = -(\ell - R\sqrt{1 + c^2R^2})\mathbf{k}$.

- **Change of basis (cylindrical basis).** $\mathcal{B}_1 = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ rotated by $\theta(t)$ about $\mathbf{k}$ ($\mathbf{k} = \mathbf{k}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Surface-adapted basis.** With $s = \sqrt{1 + 4c^2R^2}$, the unit meridian tangent is $\mathbf{e}_g = (\mathbf{e}_R + 2cR\,\mathbf{k})/s$ (proportional to $\mathbf{t}_R$), the unit normal is $\hat{\mathbf{n}} = (-2cR\,\mathbf{e}_R + \mathbf{k})/s$ (proportional to $\mathbf{n}$ above) and $\mathbf{e}_g \times \mathbf{e}_\theta = \hat{\mathbf{n}}$, so $\{\mathbf{e}_g, \mathbf{e}_\theta, \hat{\mathbf{n}}\}$ is right-handed and

$$[{}_1R_2] = \begin{pmatrix} 1/s & 0 & -2cR/s \\ 0 & 1 & 0 \\ 2cR/s & 0 & 1/s \end{pmatrix}, \qquad \det[{}_1R_2] = \frac{1+4c^2R^2}{s^2} = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

- Both determinants and products $RR^T$ were verified symbolically with sympy.

PHASE 3

## Step-by-Step Mathematical Deduction

**Part (a)-(d): Forces and Vectors**

1. Forces on $P$: Gravity $-mg\mathbf{k}$, normal reaction $\mathbf{N} = N\frac{-2cR\mathbf{e}_R+\mathbf{k}}{\sqrt{1+4c^2R^2}}$, cord tension $\mathbf{T}_P = -T\frac{\mathbf{r}_P}{|\mathbf{r}_P|}$.

2. Forces on $Q$: Gravity $-mg\mathbf{k}$, cord tension $\mathbf{T}_Q = +T\mathbf{k}$.

**Part (e): Tension Condition**

The string remains in tension as long as $T(t) > 0$.

For $Q$, $m\ddot{z}_Q = T - mg \implies T = m(g + \ddot{z}_Q) > 0 \iff \ddot{z}_Q > -g$.

**Part (f)-(g): Two First Integrals of Motion**

1. **Angular Momentum Conservation:**

The surface is axisymmetric and the cord tension is central (directed towards $O$). Neither exerts torque about $Oz$:

$$L_z = mR^2\dot{\theta} = \text{constant} = m[R(0)]^2 d = m\frac{d}{c^2}$$

$$\dot{\theta}(t) = \frac{d}{c^2R^2}$$

2. **Total Mechanical Energy Conservation:**

Constraints are frictionless and scleronomic:

$$E = T_P + T_Q + V_{g,P} + V_{g,Q} = \text{constant}$$

$$T_P = \frac{1}{2}m\left[\dot{R}^2(1 + 4c^2R^2) + R^2\dot{\theta}^2\right]$$

$$T_Q = \frac{1}{2}m\dot{z}_Q^2$$

$$V = mg(cR^2) + mgz_Q$$

**Part (h): Turning Point Equation $f(R) = 0$**

Setting $\dot{R} = 0$ in the conserved energy equation $E(R, \dot{R} = 0) = E_0$ yields the implicit algebraic equation $f(R) = 0$ whose positive roots define the minimum and maximum radii $R_{min}$ and $R_{max}$.

PHASE 4

### Physical Interpretation & Dimensional Verification

- The particle executes bounded precessing orbits on the paraboloid, bounded between two parallel circles $R_{min}$ and $R_{max}$.
- **Dimensional check:** $[c] = \mathrm{m}^{-1}$, $[d] = \mathrm{s}^{-1}$, $R(0) = 1/c$ is a length; $[cR^2] = \mathrm{m}$ and $[c\,R] = 1$, so $\sqrt{1 + c^2R^2}$ is dimensionless.
- **Explicit turning-point equation (part (h)).** At $\dot{R} = 0$ the kinetic energy reduces to $\frac{1}{2}mR^2\dot{\theta}^2$ with $R^2\dot{\theta}^2 = d^2/(c^4R^2)$ (from $L_z$), and the potential energy is $mg\,[cR^2 + R\sqrt{1 + c^2R^2}] - mg\ell$ (the term $-mg\ell$ cancels in $E - E_0$). Equating with the initial energy at $R(0) = 1/c$ (where $R\sqrt{1 + c^2R^2} = \sqrt{2}/c$), per unit mass:

$$f(R) = \frac{d^2}{2c^4R^2} + g\left(cR^2 + R\sqrt{1+c^2R^2}\right) - \frac{d^2}{2c^2} - \frac{(1+\sqrt{2})\,g}{c} = 0$$

Each term has units $\mathrm{m}^2/\mathrm{s}^2$. The allowed radii are those with $f(R) \le 0$, and $R = 1/c$ is a root (sympy: $f(1/c) = 0$).

- **Numerical example ($c = d = g = 1$):** the roots of $f$ are $R = 0.4828$ and $R = 1$ (the launch point is the outer turning point), $f(0.7) = -0.549$ and $f(0.9) = -0.276$ are negative (allowed band between the roots), and $f(0.4) = 0.802$, $f(1.2) = 0.747$ are positive (forbidden), so the orbit is bounded between $R_{min} = 0.4828/c$ and $R_{max} = 1/c$.

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*Source: Problems.pdf (Page 33)*